

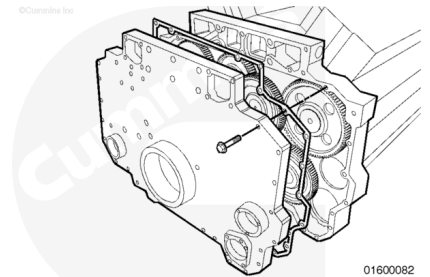
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Remove the front gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html>)

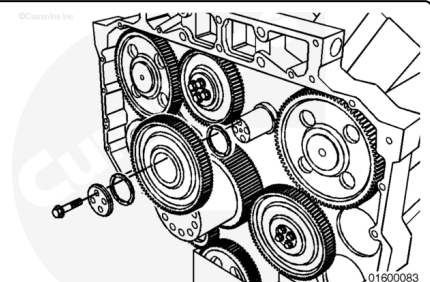


Remove

Mark the gears before disassembly so proper reassembly can occur or align the factory timing marks. Cummins Inc., recommends measuring the camshaft idler gear end clearance before removing the gear.

Remove the four mounting capscrews, thrust retainer plate, thrust bearing, and the left bank camshaft idler gear

Remove the idler shaft.



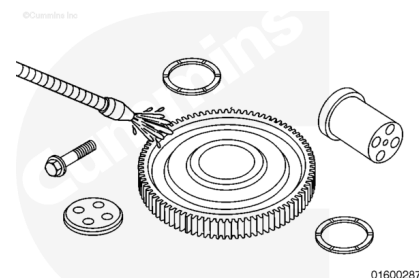
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

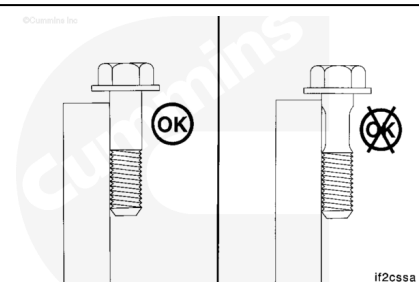


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Clean the parts with solvent and dry with compressed air.

Use a straight edge to inspect the retaining capscrews for necking.

If the retaining capscrews are necked. The retaining capscrews **must** be replaced.

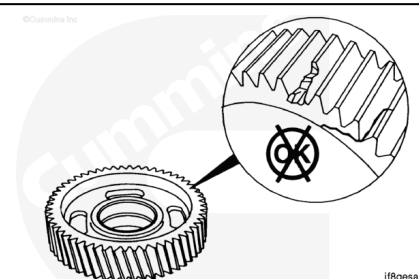


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Inspect the idler gear for chipped, broken or cracked teeth.

Inspect the bushing for damage.

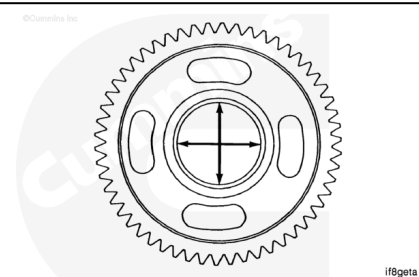
If the idler gear bushing or teeth are damaged the idler gear **must** be replaced.



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Measure the inside diameter of the idler gear bushing.

Left Bank Camshaft Idler Gear Inside Diameter		
mm		in
73.035	MIN	2.8753
73.065	MAX	2.8765



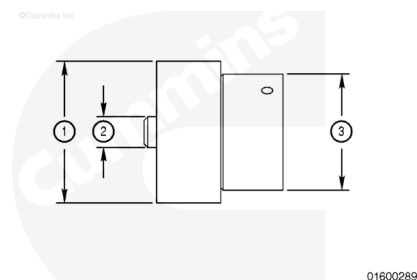
if8geta

The bushing in the idler gear is precision bored after installation. Replace the bushing and gear as an assembly if the idler gear bushing is **not** within specification.

If the idler gear is a compound gear it **must** be replaced as an assembly.

Measure the outside diameter of the idler gear shaft.

Left Bank Camshaft Idler Gear Shaft Outside Diameter			
	mm		in
(1)	88.75	MIN	3.494
	89.05	MAX	3.506
(2)	21.975	MIN	0.8652
	22.000	MAX	0.8661
(3)	72.92	MIN	2.871
	72.98	MAX	2.873



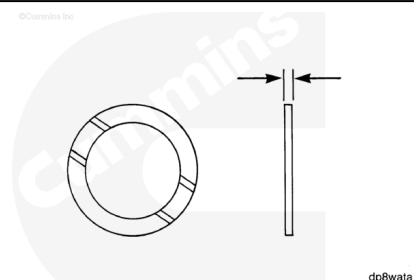
If the idler gear shaft is **not** within specifications. The idler gear shaft **must** be replaced.

Check the grooved side of the thrust bearing for damage.

If the grooved side of the thrust bearing is damaged, the thrust bearing **must** be replaced.

Measure the thrust bearing thickness.

Left Bank Camshaft Idler Gear Thrust Bearing Thickness		
mm		in
2.337	MIN	0.092
2.387	MAX	0.094



If the thrust bearing is **not** within specification. The thrust bearing **must** be replaced.

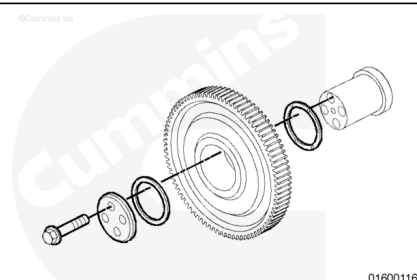
Install

The grooves in the thrust washer **must** be positioned toward the gear.

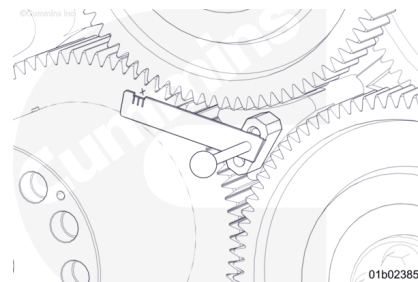
The timing marks on the camshaft idler gear **must** be visible when the gear is installed.

Use Lubriplate™ 105, or equivalent to lubricate the gear bushing, shaft, and thrust washer.

Lubricate the capscrews with clean engine oil. Assemble the parts as illustrated.



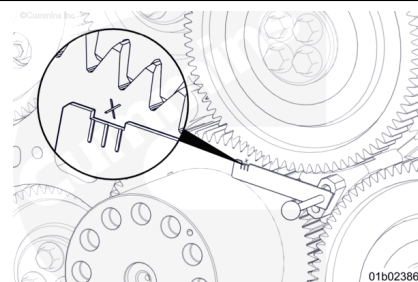
Insert the crankshaft timing service tool, Part Number 5394520, into the position shown in the front gear housing.



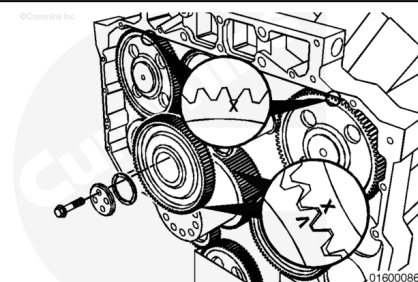
Use the crankshaft timing service tool to set the crankshaft timing mark at the Top Dead Centre (TDC) position.

Rotate the crankshaft so that the 'X' timing mark on the crankshaft gear aligns as closely as possible to the center line on the crankshaft timing tool.

Note : The 'X' timing mark on the crankshaft gear **must** fall between the two outer lines on the crankshaft timing tool.



Align the "V" on the idler gear with the "X" on the crankshaft gear. Align the "X" on the camshaft gear at 12 o'clock and with the dowel holes in the front gear housing.



Rotate the idler shaft as necessary to align the mounting capscrew holes.

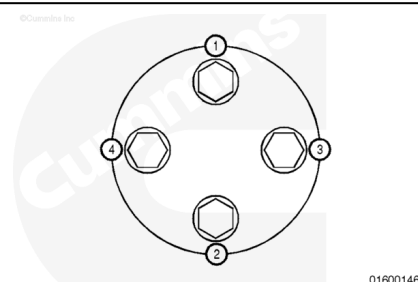
The oil holes in the idler shaft **must** be placed in the vertical position.

Use the capscrews to pull the shaft into the bore.

Using the following steps, tighten the capscrews in the sequence shown.

Torque Value:

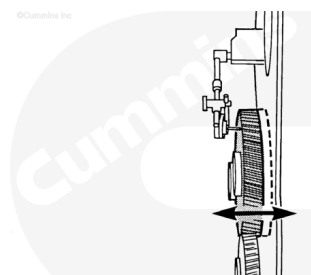
1. 80 n•m [59 ft-lb]
2. 165 n•m [122 ft-lb]
3. 280 n•m [207 ft-lb]



Measure the camshaft idler gear end clearance.

Left Bank Camshaft Idler Gear End Clearance

mm		in
0.15	MIN	0.006
0.33	MAX	0.013

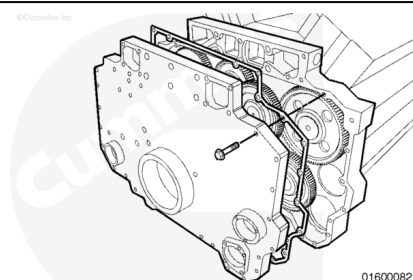


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Finishing Steps

⚠ WARNING ⚠

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- Install the front gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html>)
- Operate the engine and check for leaks.